



COPERNICUS MASTER
IN DIGITAL EARTH

Master Thesis Topic Proposal

*For Students Enrolled in the CDE Joint Programme Between
Paris-Lodron Universität Salzburg and Palacký University Olomouc*

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Mapping, monitoring and communicating landslide information
through an interactive dashboard

Proposed Thesis Title:

1. Motivation and Relevance (1000 characters max)

Landslides pose significant threats to communities, especially in vulnerable, disaster-prone regions where timely information is critical for humanitarian response. However, the complexity of Earth Observation (EO) data and the lack of accessible tools often hinder rapid action in the field. This thesis contributes to the LEONA project which aims to develop user-oriented landslide hazard information services to support humanitarian aid organizations like Médecins Sans Frontières (MSF) by designing and prototyping an interactive dashboard that visualizes landslide data in an operationally useful way. The tool bridges the gap between EO-based landslide mapping and field-level humanitarian decision-making, aligning with SDGs 11 and 13. The dashboard will be tested using case studies from high-risk regions like Mozambique, Madagascar, Malawi and Colombia.

2. Research Questions (1000 characters max)

- How can EO-based landslide information be effectively visualized and communicated in a dashboard format for humanitarian users like MSF?
- What are the optimal design elements and visual encodings (e.g., time sliders, color schemes, alerts) for communicating landslide risk and impact?
- How can multi-source EO data (e.g., Sentinel-1 SAR, Sentinel-2, DEM, rainfall) be integrated and filtered for different regions?
- How can user requirements from humanitarian actors help the dashboard's functionalities and usability?

3. Methodology and Data (1000 characters max)

The dashboard will be built using open-source web GIS tools (e.g., Leaflet.js, Dash/Plotly, or OpenLayers) and will integrate multiple data sources: Sentinel-1 and Sentinel-2, precipitation data (e.g., CHIRPS or GPM), digital elevation models, and historical landslide inventories. The design process will follow a user-centered methodology, incorporating MSF and LEONA feedback loops. Initial deployment and testing will focus on LEONA's study regions—Malawi, Madagascar, Mozambique, and Colombia. Backend processing may use Python and Google Earth Engine for automation. Iterative usability evaluations will help refine the interface and content delivery.

4. Expected Outcomes and Contributions (500 characters max)

An interactive, prototype dashboard that visualizes EO-based landslide hazard data for humanitarian purposes. It will support timely communication, improve situational awareness, and offer recommendations for operational integration within MSF workflows.

5. Risks and Dependencies (1000 characters max)

Challenges may include inconsistent EO data availability (e.g., cloud cover for optical), limited access to real-time ground truth for validation, and delayed user feedback from humanitarian partners. To mitigate this, historical data will be used for testing, and a minimum viable product will be developed early for internal feedback. The dashboard will be designed to be modular and lightweight to ensure compatibility in low-bandwidth humanitarian settings. Dependency on external APIs (e.g., Copernicus, GEE) will be monitored and documented for future scalability.

References

- GFDRR/ThinkHazard platform
- UNOSAT Map Viewer
- Petley et al. (2022) – landslide EO monitoring methods
- Sepúlveda & Petley (2015) – Landslide trends in Latin America
- Open-source EO APIs: Sentinel Hub, GEE, Copernicus Data Space Ecosystem

Supervisor Statement:

(This section is for supervisors to fill in after reviewing the proposal)

Comments:

Decision: *(Highlight your choice)*

- ☐ Accepted
- ☐ Rejected
- ☐ Subject to major revision